**Problem Set 3**
Intermediate Microeconomics
Spring 2023

1. There is currently a sales tax on all cars, foreign and domestic. In order to help the Taiwanese car industry, the government is thinking of eliminating the sales tax. Plan A is to eliminate the tax for domestic cars only; Plan B is to eliminate the tax for both domestic and foreign cars.

   a. Which plan is better for domestic car makers?

   b. **True or False**: Plan B, combined with an appropriate redistribution of income, can make everybody happier than Plan A. If your answer is true, explicitly describe the appropriate redistribution of income. If your answer is false, explain carefully why no such redistribution is possible.

2. Consider an exchange economy with two goods, $x$ and $y$, and two consumers, Ring and Ting. Ring's utility function us $u_r=x_ry_r$, and her endowment is $w_r=(2,2)$. Ting's utility function is $u_t=x_ty_t^2$ and her endowment is $w_t=(3,3)$. Let the price of $x$ equal to 1. Find the competitive equilibrium. That is, solve for the market-clearing price, $p_y$, and consumption, $(x_r^*, y_r^*)$ and $(x_t^*, y_t^*)$ .

3. There are two goods in the world, milk ($x$) and honey ($y$), and two consumers, Mike and Shepard. Mike's utility function is $u_m=x_my_m^3$ and his endowment is $w_m=(4,4)$. Shepard's utility function is $u_s=x_sy_s$ and his endowment is $w_s=(0, 0)$. We will again let $p_x=1$ and let $p_y$ vary.

   a. Is the original endowment Pareto efficient?

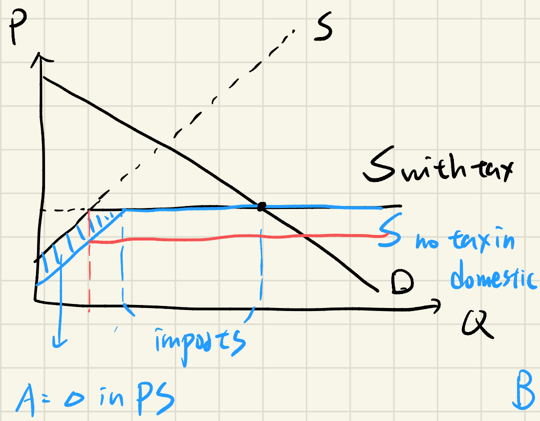
   Suppose the dictator sets Mike's lump-sum tax at $T_m=-4$, and Shepard's lump-sum transfer at $T_s=4$.

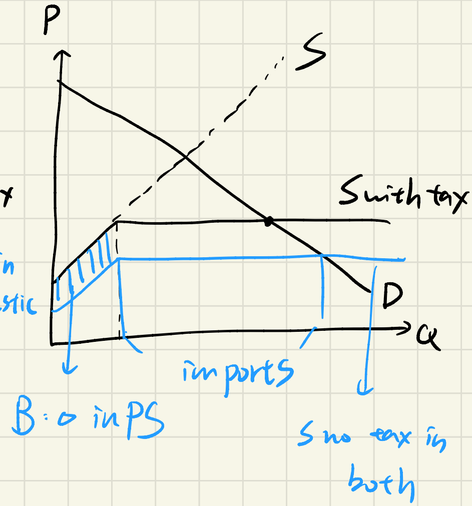
   b. Solve for the competitive equilibrium.

   c. Prove that the new equilibrium allocation is Pareto efficient.

1.

a.

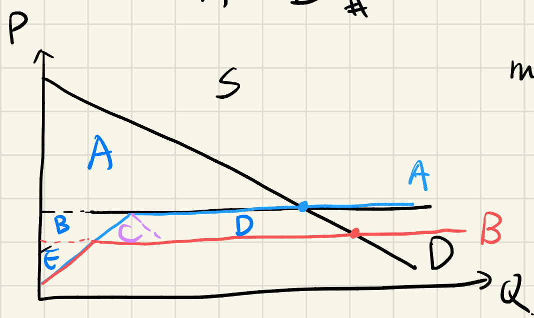



A > B #

more from A to B

b.




| | A | B | o | |
|---|---|---|---|---|
| CS | A | A+B+C+D | B+C+D | C < B+D |
| PS | B+E | E | -B | |
| T | D | 0 | -D | |
| W | | | C | |

not possible to make everyone happier

2. Consider an exchange economy with two goods, $x$ and $y$, and two consumers, Ring and Ting. Ring's utility function us $u_r=x_r y_r$, and her endowment is $w_r=(2,2)$. Ting's utility function is $u_t=x_t y_t^2$ and her endowment is $w_t=(3,3)$. Let the price of $x$ equal to 1. Find the competitive equilibrium. That is, solve for the market-clearing price, $p_y$, and consumption, $(x_r^*, y_r^*)$ and $(x_t^*, y_t^*)$ .

$$P_x = 1$$

$$U_r = x_r y_r \quad W_r(2,2) \qquad U_t = x_t y_t^2 \quad W_t(3,3) \qquad P_y \text{ vary}$$

$$x_r + x_t = 5 \qquad I_r = P_x \cdot x_r + P_y \cdot y_r = 2 + 2P_y$$

$$y_r + y_t = 5 \qquad I_t = P_x \cdot x_t + P_y \cdot y_t = 3 + 3P_y$$

$$x_r = \frac{1}{2} \cdot \frac{2+2P_y}{P_x} \qquad y_r = \frac{1}{2} \cdot \frac{2+2P_y}{P_y}$$

$$x_t = \frac{1}{3} \cdot \frac{3+3P_y}{P_x} \qquad y_t = \frac{2}{3} \cdot \frac{3+3P_y}{P_y}$$

$$\Rightarrow \frac{1}{2} \cdot \frac{2+2P_y}{\cancel{P_x}\,1} + \frac{1}{3} \cdot \frac{3+3P_y}{\cancel{P_x}\,1} = 5$$

$$1 + P_y + 1 + P_y = 5 \Rightarrow 2P_y = 3 \Rightarrow P_y = 1.5$$

$$\Rightarrow \frac{1}{2} + \frac{2+2P_y}{P_y} + \frac{2}{3} \cdot \frac{3+3P_y}{P_y} = 5$$

$$\frac{1}{P_y} + 1 + \frac{2}{P_y} + 2 = 5 \Rightarrow \frac{3}{P_y} = 2 \Rightarrow P_y = 1.5$$

$$(x_r^*, y_r^*) = \left(\frac{5}{2}, \frac{5}{3}\right) \qquad (x_t^*, y_t^*) = \left(\frac{5}{2}, \frac{10}{3}\right)$$

3. There are two goods in the world, milk ($x$) and honey ($y$), and two consumers, Mike and Shepard. Mike's utility function is $u_m=x_m y_m^3$ and his endowment is $w_m=(4,4)$. Shepard's utility function is $u_s=x_s y_s$ and his endowment is $w_s=(0, 0)$. We will again let $p_x=1$ and let $p_y$ vary.

a. Is the original endowment Pareto efficient?

Suppose the dictator sets Mike's lump-sum tax at $T_m=-4$, and Shepard's lump-sum transfer at $T_s=4$.

b. Solve for the competitive equilibrium.

c. Prove that the new equilibrium allocation is Pareto efficient.

$U_m = x_m y_m^3 \quad w_m(4,4)$

$U_s = x_s y_s \quad w_s(0,0)$

$x_m + x_s = 4 \quad P_x = 1$

$y_m + y_s = 4 \quad P_y$ vary

a. It is P.E. Shepard cannot be better off without making Mike worse off, there is no move from the orginal endowment

$I_m = 4 + 4P_y - 4 \quad I_s = 4$

$x_m = \frac{1}{4} \cdot \frac{\cancel{4}+4P_y\cancel{-4}}{1} \quad y_m = \frac{3}{4} \cdot \frac{\cancel{4}+4P_y\cancel{-4}}{P_y}$

$x_s = \frac{1}{2} \cdot \frac{4}{1} \quad y_s = \frac{1}{2} \cdot \frac{4}{P_y}$

$\Rightarrow \frac{1}{4} \cdot \frac{4P_y}{1} + 2 = 4 \Rightarrow P_y = 2$

$(x_m^*, y_m^*) = (2, 3) \quad (x_s^*, y_s^*) = (2, 1)$

c. $MRS_m = \frac{y_m^{\cancel{3}} y_m}{3 x_m \cancel{y_m^2}} = \frac{3}{3 \times 2} = \frac{1}{2}$

$MRS_s = \frac{y_s}{x_s} = \frac{1}{2}$

$MRS_m = MRS_s = \frac{1}{2} \Rightarrow$ new eqm is P.E.